

PAPER_671: UQFF dM/dt Derivation

Subtitle: Full step-by-step derivation of the UQFF-modified mass loss rate, with analytic $M(t)$ approximation and Euler simulation. **Module:** UQFFDMDtDerivation

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Abstract

We present the complete three-step UQFF derivation of the modified black hole mass loss rate dM/dt , showing a $\sim 33\times$ suppression relative to standard Hawking emission.

1. Standard Hawking Rate

$$\left. \frac{dM}{dt} \right|_{std} = -\frac{L_H}{c^2} = -\frac{\hbar c^4}{15360\pi G^2 M^2 c^2}$$

2. UQFF Step 1 — f_{TRZ}

$$\frac{dM'}{dt} = \left. \frac{dM}{dt} \right|_{std} \cdot (1 - f_{TRZ})$$

Physical basis: negentropic restoring force suppresses pair production at horizon.

3. UQFF Step 2 — $[UA]/[SCm]$

$$\frac{dM''}{dt} = \frac{dM'}{dt} \cdot \frac{\rho_{SCm}}{\rho_{UA}}$$

4. UQFF Step 3 — U_m Anchor

$$\left. \frac{dM}{dt} \right|_{UQFF} = \frac{dM''}{dt} \cdot \exp\left(-\frac{U_m}{k_B T_H}\right)$$

5. Combined Formula

$$\left. \frac{dM}{dt} \right|_{UQFF} = \left. \frac{dM}{dt} \right|_{std} \cdot (1 - f_{TRZ}) \cdot \frac{\rho_{SCm}}{\rho_{UA}} \cdot \exp\left(-\frac{U_m}{k_B T_H}\right)$$

Suppression factor: $0.9 \times 0.1 \times e^{-1} \approx \mathbf{0.033}$ (evaporation $\sim 30\times$ slower).

6. Analytic Approximation

$$M(t) \approx (M_0^3 - 3A t)^{1/3}$$

where $A = |dM/dt|_{UQFF} \times M_0^2$.

7. C++ Module

UQFFDMDtDerivation.h / .cpp — Session 172 CP4 #255 — UQFFDMDtDerivationCalculator

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